COSC2753 – Machine Learning

Assignment 1: Life Expectancy Modelling

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I declare that in submitting all work for this assessment I have read, understood and agreed to the content and expectations of the Assessment declaration.

I. Introduction

- **Life Expectancy Metrics Overview:** Life expectancy is a widely used metric to evaluate the country's health by estimating how long a person is expected to live based on current mortality trends. It can contribute to strategic decisions in healthcare, economic planning, and policy development.
- **Ultimate Goal:** This project aims to develop an accurate, efficient, and generalizable model to predict life expectancy for unseen countries.
- **Scope:** This report outlines the complete modeling pipeline, including Data Inspection, Cleaning, Exploratory Analysis, Feature Engineering, Model Training, and Ultimate Judgement.

II. Dataset Overview

- **Size**: The dataset contains 2,071 rows and 23 columns, capturing country-level data across years. Each row represents a unique combination of a country and a year.
- **Column types**: "Country", "Year", and "Status" are numerical categories, while the rest are continuous variables.
- **Duplication**: No duplicated rows were found in the dataset.
- **Missing Values**: 65 rows contain missing values in the dataset. Columns with the most missing values are: "Total expenditure", "Percentage expenditure", and "GDP".

III. Missing Values and Outliers Handling

A. Column-wise Inspection:

- **Missing Values**: Mostly located in "Total expenditure", "Percentage expenditure", "GDP", "BMI", "Population", "Schooling", and "IncomeCompositionOfResources".
- **Column-wise Analysis**: 15 features with skewness and outliers due to the diversity across different countries around the world.
- **Implication of column-wise**: These outliers lack context, as the abnormal value globally may be normal locally. Hence, it raises the need for country-wise analysis.

B. Country-wise Inspection

- **Time-Series Context:** If outliers exist, they should reflect within each country's trend over the time series rather than column-wise.
- **Random Country Inspection**: The BMI of country 84 has gradually increased but showed unusual values in 2015 and 2016. Similar patterns occur in other features such as Adult Mortality (General, Male, and Female).

C. Imputation

1. Approaches

- Country-wise outlier marking using Interquartile Range (IQR).
- MultiOutputRegressor Imputation for multicollinear feature blocks
- KNNImputer for remaining values except for Mode-filling for Status

2. Multicollinearity Detection

- Utilize hierarchical clustering on numerical features with Pearson correlation as the distance metric. 4 clusters were identified.

3. Imputation pipeline

- **Outliers Marking**: Mark as NaN, and some features need to remain stable, while others can tolerate extreme outliers like an epidemic at a particular time.
- **Block-Wise Imputation (MultiOutputRegressor):**
  - **Blocks**: Each is a cluster of multicollinearity features identified above.
  - **Method**: For each country, train a separate model on complete rows, then predict the missing values for each block to preserve the dataset nature.
- **Remaining Values**:
  - Country-wise KNN Imputer for numerical features
  - Mode-filling for “Status”

4. Post-Imputation Check

- Country-wise anomalies were resolved and clusters preserved, resulting in smoother time-series trends while retaining the dataset's nature (**Figure 1**).

IV. Exploratory Data Analysis

A. Univariate Feature Analysis

- **Target variable** (Figure 2): Slightly left-skewed distribution, most values between 63 and 76 years. Lower-end outliers reflect countries with poor health status.
- **Others**: Most predictors exhibit skewness and scale imbalance. Outliers are natural due to global diversity and it is acceptable as local true outliers were imputed above.

B. Target-Feature Analysis

- **Strong Negative Correlation**: Adult Mortality (General, Male, Female) (Figure 3), “HIV-AIDS”, “Thinness1-19years”, “Thinness5-9years” → Main factors that reduce life expectancy as they are disease and mortality metrics.
- **Strong Positive Correlation**: “IncomeCompositionOfResources”, “BMI” (Figure 3), “Schooling”, “Polio”, “Diphtheria” → Positively affect the target variable due to protective health metrics and socioeconomic factors.
- **Problematic Features**: “Population” correlates only 0.03 with the target, “Country” is just an unnecessary local-level identifier, and “Year” might cause overfitting.

C. Feature–Feature Relationship

- **High Correlation** ($r > 0.90$): “AdultMortality-Male” & “AdultMortality-Female”, “Thinness1-19years” & “Thinness5-9years”, “SLS” & “Under5LS”, “Polio” & “Diphtheria”, and “GDP” & “PercentageExpenditure”.
- **Actions**: Combine or remove to avoid redundancy and multicollinearity.

D. EDA-Based Recommendation

- **Drop**: “AdultMortality-Male”, “AdultMortality-Female”, “GDP”, “Population”, “Country”, “Year” to reduce noise and overfitting risk.
- **Combined Pair:**
  - “Polio” + “Diphtheria” → “VaccineScore”
  - “Thinness1-19” + “Thinness5-9” → “ThinnessIndex”
  - “SLS” + “Under5LS” → “EarlyMortality”

- **Cross-Validation Strategy**: Grouped Cross-Validation by “Country” to simulate the real-world pattern of predicting data from unseen countries.
- **Model Selection Strategy**:
  - Prioritize tree-based models (e.g., Random Forest, XGBoost) for nonlinear adaptability and strong generalization.
  - Avoid linear models and distance models like KNN, due to poor cross-country adaptability.

V. Feature Engineering

A. Feature Preprocessing

- **Train-Test Split by Country**: The dataset was split using 80% of countries for training and 20% for validating. This ensures the clear separation between train and test and promotes generalization to unseen nations.
- **Feature Cleaning**: Dropped redundant columns as identified in the EDA phase: “Country”, “Year”, “GDP”, “Population”, “AdultMortality-Male”, “AdultMortality-Female”
- **Feature Combination**: three new columns were constructed:
  - **VaccineScore** = Mean of “Polio” and “Diphtheria”.
  - **ThinnessIndex** = Mean of “Thinness1-19years” and “Thinness5-9years”.
  - **EarlyMortality** = Mean of “SLS” and “Under5LS”.
- **Feature Transformation**: Applied log1p transformation to right-skewed features to balance the distribution and somewhat scale down the feature.
- **Feature Scaling**: Standardized all numerical features except for “Status” and “TARGET_LifeExpectancy” using “StandardScaler” to ensure zero mean and unit variance across features.

B. Feature Selection

In this section, several methods will serve as a tool to ensure the currently selected set rather than using it to remove some features.

1. Variance Threshold:

- **Methodology**: Detects and removes features with low variability across samples.
- **Result**: All 13 features passed, indicating all variables provide meaningful variation.

2. Pearson Correlation

- **Methodology**: Evaluate the linear relationship between each feature and the target variable.
- **Result**: All 13 features passed, showing sufficient correlation strength.

3. XGBoost Feature Importance

- **Methodology**: Train a XGBoost model to automatically assign importance scores.
- **Result**: All 13 features passed, meaning each of them is important to predictive performance

4. Conclusion

- No features were dropped in any method, confirming the selection from the EDA phase for the feature set is appropriate for modeling.

VI. Modeling

A. Setup

- **Tree-based Configuration**: GradientBoosting, HistGradientBoosting, ExtraTrees, XGBoost, RandomForest, LightGBM, AdaBoost. As stated in the EDA, these are the priority of this project due to their robustness in capturing country-level patterns and non-linear relationships that frequently appear in the given dataset.
- **Linear-based Configuration**: LinearRegression in base form, Ridge and ElasticNet regularization. They are just for comparison purposes only because they try to use the global coefficient to predict local patterns.
- **Distance-based Configuration**: K Nearest Neighbor (KNN). This model is also for comparison purposes because this model is also not suitable for the dataset that has a lot of local patterns.
- **Country-wise Cross Validation**: To mimic real-world generalization, GroupKFold was used with grouping by Country to ensure that no country appears in both training and validation folds, preventing data leakage and providing robust performance evaluation.

B. Tuning

- **Methodology**: Each model was tuned using RandomizedSearchCV with 5-fold GroupKFold and 50 iterations. R^2 and tuning time was recorded to observe not only raw performance but also the time tradeoffs of the model.
- **Top Performer**: The top 6 performances are from ensemble tree-based models. It is notable that Gradient Boosting achieved the highest R^2 (0.845) while HistGradientBoosting is slightly less accurate (0.843) but the tuning time is 6 times lower than GradientBoosting (~77s vs ~512s)
- **Mid-Tier Performer**: The R^2 of KNN is approximately 0.81, showing a weaker generalization of distance-based models to unseen countries. However, KNN's tuning time is just roughly 7.5 seconds.
- **Underperform**: Although the tuning time for linear models is very fast (~0.13s without regularization and ~1.3s with regularization), these models show a significant gap in terms of accuracy, as their R^2 is just between 0.74 - 0.75.
- **Insight**: This result verifies the earlier EDA-driven hypothesis that tree-based learners are more effective at capturing country-level variations, and the trade-off for best performance or best efficiency is notable.
- See Figure 4 for full training results.

C. Validation

- **Top Performer**:
  - All six top models are tree-based ensembles,
  - Achieved MAEs ~2.5 years, acceptable for life expectancy prediction,
  - Residuals show no major bias, predictions align well with the actual value
  - **Best accuracy**: XGBoost (R^2 = 0.893, training: 185s, inference: 0.008s)
  - **Best efficiency**: HistGradientBoosting with competitive accuracy (R^2 = 0.875), excellent speed (88s training, 0.004s inference)

- **Mid-Tier Models**:
  - AdaBoost (R^2 = 0.831) and KNN (R^2 = 0.817) show moderate performance.
  - Confirms distance-based methods like KNN are less suitable for generalization across countries as stated in the EDA.
- **Underperformers:**
  - Linear models ($R^2 \approx 0.76$) underfit and fail to capture nonlinear country-level patterns, once again matching with the EDA hypothesis.
  - Fast but not suitable to capture complex patterns in the given dataset.
- **Insights**:
  - **Avoid overfitting**: The validation trend is similar to the training trend → Confirm the strength of the column-wise strategy to avoid overfitting while ensuring accuracy.
  - **Hypothesis confirmation**: The validation result confirms the earlier EDA hypothesis for model selection and performance.
- See Figure 4 for full validation results.

VII. Ultimate Judgement

A. Candidates Comparison

- **XGBoost**:
  - **Best Performance**: R^2 = 0.893 (highest), RMSE = 3.08, MAE = 2.42,
  - **Slower Speed**: training: 185s, inference: 0.008s,
  - **Visualization**: Clean, symmetric residuals, strong fit to actual values (Figure 5)
- **Hist Gradient Boosting**:
  - **Competitive Performance**: R^2 = 0.875, RMSE = 3.33, MAE = 2.56,
  - **Much Faster Speed**: training: 77s, inference: 0.004s,
  - **Visualization**: Visual diagnostics are almost similar to XGBoost, suggesting comparable generalization quality (Figure 6).

B. Final Decision

- **Hist Gradient Boosting is chosen as the final model** because:
  - **Competitive Performance**: R^2 = 0.875 - only 1.7% difference toward XGBoost.
  - **Time Efficiency**: 2x speedup in comparison with XGBoost.
  - **Most Balance**: Ensure performance and speed, allowing further iterative retraining or real-time inference.
  - **Plotting**: Visual behavior and generalization are almost similar to XGBoost.

C. Model Limitation

- Slightly lower performance than XGBoost
- Limited interpretability compared to linear models

D. Limitation Handling

- **Tuning**: Apply exhaustive grid search if higher accuracy is required
- **Interpretability**: Use SHAP to interpret predictions and feature impact

Figure

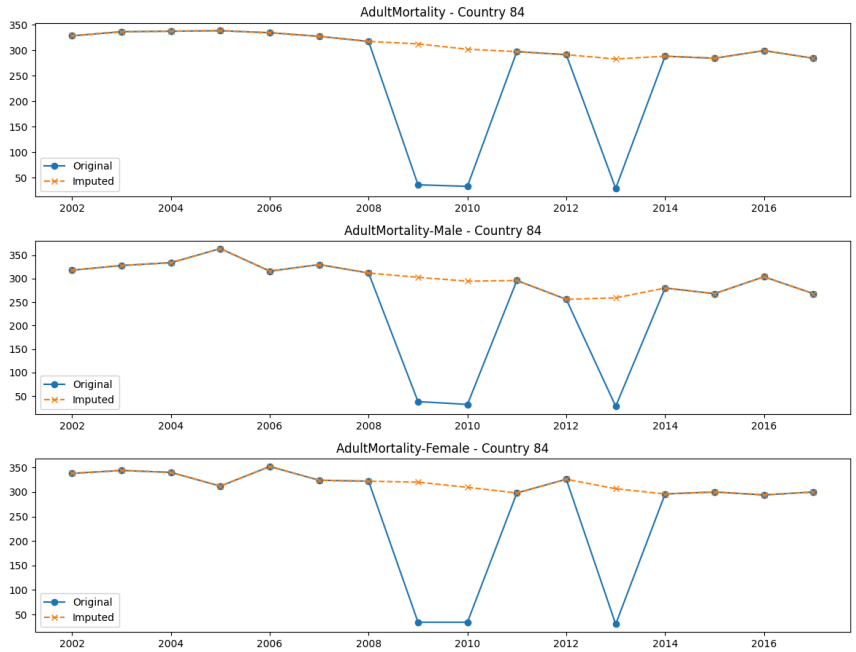

Figure 1: AdultMortality feature before and after imputation

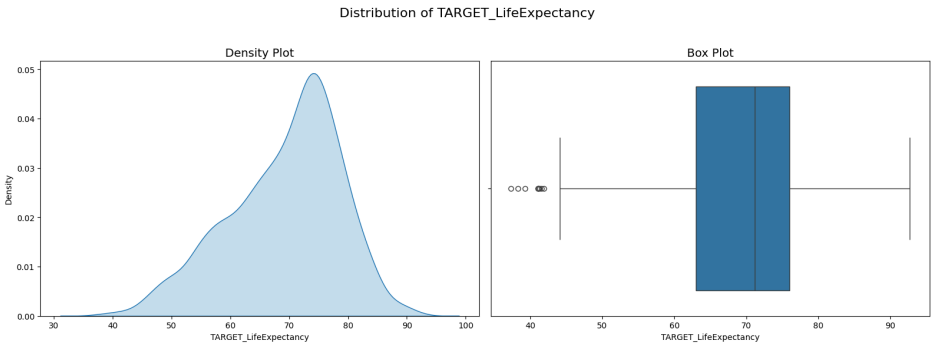

Figure 2: TARGET_LifeExpectancy distributions

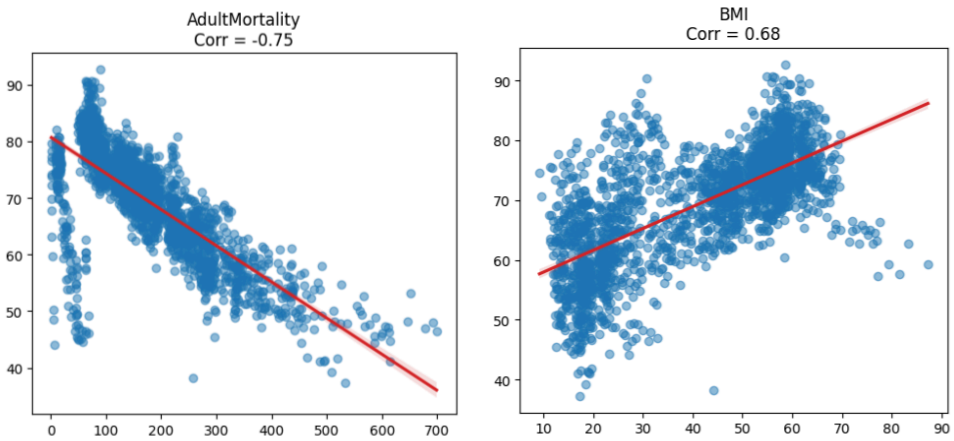

Figure 3: Visualization of relationships between AdultMortality and BMI to the target

Training Summary:

	Best R^2 (CV)	Total Search Time (sec)
GradientBoosting	0.844779	512.987504
XGBoost	0.844253	185.01188
HistGradientBoosting	0.843216	76.776488
ExtraTrees	0.840611	130.190825
RandomForest	0.839902	364.346157
LightGBM	0.828318	155.634007
KNN	0.809986	7.477787
AdaBoost	0.808601	115.840908
ElasticNet_Linear_Regression	0.752809	1.29826
Ridge_Linear_Regression	0.750116	1.433602
Linear_Regression	0.739441	0.133011

Validation Summary:

	R^2	RMSE	MSE	MAE	MedAE	MAPE	Inference Time (Seconds)
XGBoost	0.893047	3.084762	9.515756	2.421342	2.050700	0.036340	0.007720
GradientBoosting	0.883680	3.217006	10.349125	2.501045	2.040616	0.037423	0.002889
ExtraTrees	0.879634	3.272481	10.709131	2.562231	2.070111	0.038415	0.056334
RandomForest	0.877380	3.302979	10.909673	2.605017	2.145312	0.039039	0.017994
HistGradientBoosting	0.875448	3.328900	11.081572	2.555180	2.154114	0.038350	0.003780
LightGBM	0.873891	3.349636	11.220063	2.549007	1.995732	0.038293	0.013253
AdaBoost	0.831125	3.876211	15.025008	3.057790	2.470451	0.046608	0.012624
KNN	0.817151	4.033403	16.268344	3.157118	2.536215	0.048449	0.024331
Linear_Regression	0.762540	4.596421	21.127089	3.573035	2.818426	0.053814	0.001329
Ridge_Linear_Regression	0.760269	4.618347	21.329132	3.582712	2.778399	0.054033	0.001300
ElasticNet_Linear_Regression	0.758765	4.632820	21.463019	3.596850	2.805960	0.054343	0.001161

Figure 4: Training and Validation Results

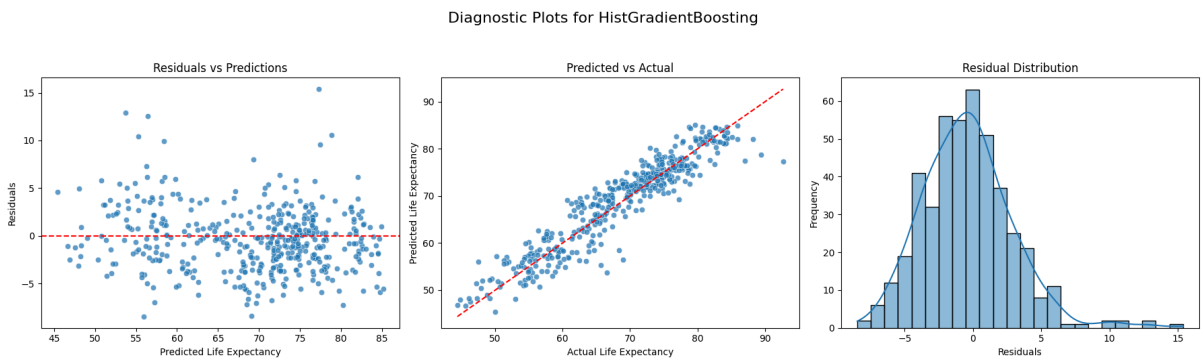

Figure 5: Diagnostic Plots for tuned Hist Gradient Boosting model

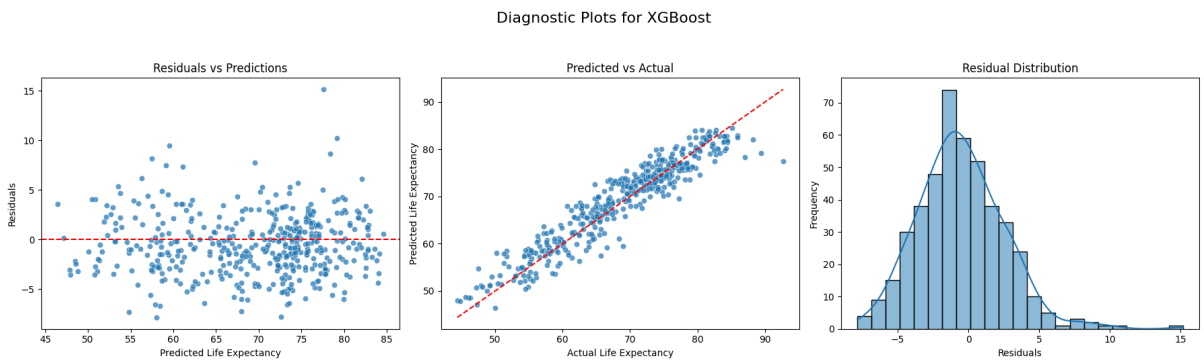

Figure 6: Diagnostic Plots for tuned Hist Gradient Boosting model